



Answers & Solutions:

1. C. Motion of charged particles

2. A. 0.25 N

Solution:

$$E = B\ell v$$

$$B = \frac{E}{\ell v} = \frac{2.5}{(0.5)(5)} = 1 \text{ T}$$

$$I = \frac{E}{R} = \frac{2.5}{5} = 0.5 \text{ A}$$

$$F = BI\ell = (1)(0.5)(0.5)$$

$$F = 0.25 \text{ N} \rightarrow \text{Ans.}$$

where:

$$E = 2.5 \text{ V}$$

$$v = 5 \text{ m/s}$$

$$\ell = 500 \text{ mm} = 0.5 \text{ m}$$

3. D. Both B and C.

4. C. Produces a fluctuating electric field.

5. A. 0.135 V

Solution:

$$E = B\ell v$$

$$E = (1.2)(0.15)(0.75)$$

$$E = 0.135 \text{ V} \rightarrow \text{Ans.}$$

where:

$$B = 1.2 \text{ T}$$

$$\ell = 15 \text{ cm} = 0.15 \text{ m}$$

$$v = 750 \text{ mm/s} = 0.75 \text{ m/s}$$

6. A. downwards

7. A. directly as the number of its turns

8. B. 0.16 A

Solution:

$$E = B\ell v = 1.2 \times 0.25 \times 8$$

$$E = 2.4 \text{ V}$$

$$I = \frac{E}{R} = \frac{2.4}{15} = 0.16 \text{ A} \rightarrow \text{Ans.}$$

9. A. attract each other

10. A. 1.56 mV

Solution:

$$E = B\ell v = (40 \times 10^{-6})(1.76) \left(\frac{80 \times 10^3}{60 \times 60} \right)$$

$$E = 1.56 \text{ mV} \rightarrow \text{Ans.}$$

11. A. 4 A

Solution:

$$F = BI\ell \sin \theta$$

$$I = \frac{F}{B\ell \sin \theta} = \frac{1.20}{1.25 \times 240 \times 10^{-3} \times \sin 90^\circ}$$

$$I = 4 \text{ A} \rightarrow \text{Ans.}$$

12. D. Both B and C.

13. A. towards the viewer

14. C. 0.582 N

Solution:

When conductor and field are at right angles,

$$F = BI\ell \sin \theta = BI\ell = \frac{\Phi}{A} I\ell$$

$$F = \frac{0.75 \times 10^{-3}}{\pi (40)^2 10^{-6}} \times 13 \times 300 \times 10^{-3}$$

$$F = 0.582 \text{ N} \rightarrow \text{Ans.}$$

where:

$$B = \frac{\Phi}{A}$$

$$\sin \theta = \sin 90^\circ = 1$$

15. A. $8.0 \times 10^{-19} \text{ N}$

Solution:

$$F = QvB = (2 \times 10^{-18})(2 \times 10^6)(2 \times 10^{-7})$$

$$F = 8 \times 10^{-19} \text{ N} \rightarrow \text{Ans.}$$

16. C. electron

17. D. Both B and C.

18. D. Both B and C.

19. C. neutrons and protons.

20. A. 1 C/s

21. C. the motion of positive charges in the opposition direction of electron flow.

22. A. electric charge.

23. D. 0.80 T

Solution:

$$F = BI\ell \sin \theta$$

$$B = \frac{F}{I\ell \sin \theta} = \frac{3.6}{15 \times 0.30 \times \sin 90^\circ}$$

$$B = 0.80 \text{ T} \rightarrow \text{Ans.}$$

24. D. Quantity of electricity

25. B. 800 s

Solution:

$$Q = It$$

$$t = \frac{Q}{I} = \frac{80}{100 \times 10^{-3}}$$

$$t = 800 \text{ s} \rightarrow \text{Ans.}$$

26. D. Both B and C.

27. D. 3600 C

Solution:

$$Q = It = (6)(10 \times 60)$$

$$Q = 3600 \text{ C} \rightarrow \text{Ans.}$$

28. B. difference in potential between 2 points

29. C. 10^6 m/s

Solution:

$$F = QvB$$

$$v = \frac{F}{QB} = \frac{10^{-20}}{(10^{-19})(10^{-7})}$$

$$v = 10^6 \text{ m/s} \rightarrow \text{Ans.}$$

30. C. The remanence of a ferrous material is measured in ampere/meter



31. **A. 21.0 N**

Solution:

$$F = BIl \sin \theta$$

$$F = 1.5 \times 70 \times 200 \times 10^{-3} \sin 90^\circ$$

$$F = 21.0 \text{ N} \rightarrow \text{Ans.}$$

32. **A. increase the reluctance**

33. **C. away from the viewer**

34. **C. an electric current**

35. **D. 25 m/s**

Solution:

$$E = B/v$$

$$v = \frac{E}{B\ell} = \frac{1.8}{(0.6)(0.12)}$$

$$v = 25 \text{ m/s} \rightarrow \text{Ans.}$$

36. **B. proton**

37. **A. towards the viewer**

38. **D. an electromagnet**

39. **A. decrease a current in a circuit**

40. **B. concentric circles**

41. **A. There is no effect**

42. **A. 5 s**

Solution:

$$Q = It$$

$$t = \frac{Q}{I} = \frac{50}{10}$$

$$t = 5 \text{ s} \rightarrow \text{Ans.}$$

43. **B. The number of protons.**

44. **D. Might have either a positive or negative charge.**

45. **A. in opposite direction**

46. **D. Is an extremely large number of charge carriers.**

47. **D. 0.1 A**

48. **C. The gauss**

49. **A. Concentrates magnetic flux lines within itself.**

50. **B. Electrostatic force.**

END